# Title

**House Price Prediction using Linear Regression**

# Aim

To develop a House Price Prediction System using the Linear Regression algorithm and evaluate its performance using the $R^2$ Score.

# Software Requirements

- Python 3.x
- Visual Studio Code / Jupyter Notebook
- Scikit-learn
- Pandas

# Dataset Used

The California Housing Dataset provided by the Scikit-learn library was used. The dataset contains various housing-related features such as median income, house age, average rooms, average bedrooms, population, latitude, and longitude. The target variable is the median house value.

# Python Code

```python
import pandas as pd
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

housing = fetch_california_housing(as_frame=True)
df = housing.frame

X = df.drop("MedHouseVal", axis=1)
y = df["MedHouseVal"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```python
print("R² Score:", r2_score(y_test, y_pred))

result = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})

print(result.head(10))
```

# Output

```
R² Score: 0.5757877060324507
   Actual  Predicted
0  0.47700   0.719123
1  0.45800   1.764017
2  5.00001   2.709659
3  2.18600   2.838926
4  2.78000   2.604657
5  1.58700   2.011754
6  1.98200   2.645500
7  1.57500   2.168755
8  3.40000   2.740746
9  4.46600   3.915615
```

# Result

The Linear Regression model was successfully trained using the California Housing Dataset. The model predicted house prices on the test dataset and achieved an $R^2$ Score of **0.5757777060**, indicating its prediction performance.